EXPERIMENT - 9

9. Implement the image scaling techniques to resize the images to both bigger and small sizes

Aim

To implement image scaling techniques in Python to resize an image into both larger and smaller sizes using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image using cv2.imread().
3. Resize the image to a larger size using cv2.resize().
4. Resize the image to a smaller size using cv2.resize().
5. Display the original, enlarged, and reduced images.
6. Close all image windows.

Program

```
import cv2

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    big = cv2.resize(img, (600, 600))
    small = cv2.resize(img, (200, 200))

    cv2.imshow("Original Image", img)
    cv2.imshow("Bigger Image", big)
    cv2.imshow("Smaller Image", small)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

Sample Input

Original image size: 400 × 400 pixels

Bigger image size: 600 × 600 pixels

Smaller image size: 200 × 200 pixels

Input image: input.jpg



Sample Output

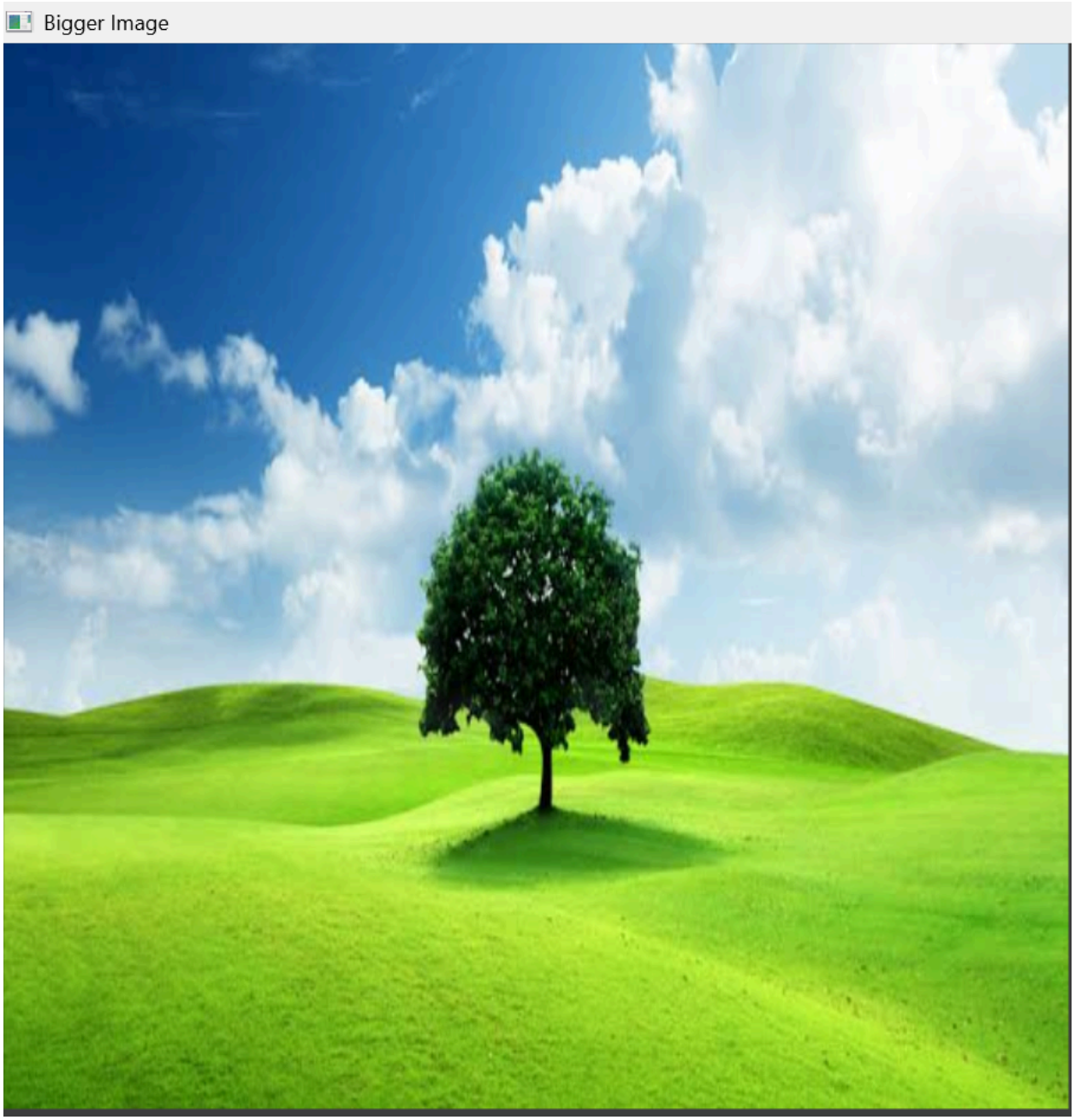
Original Image

400 × 400 pixels



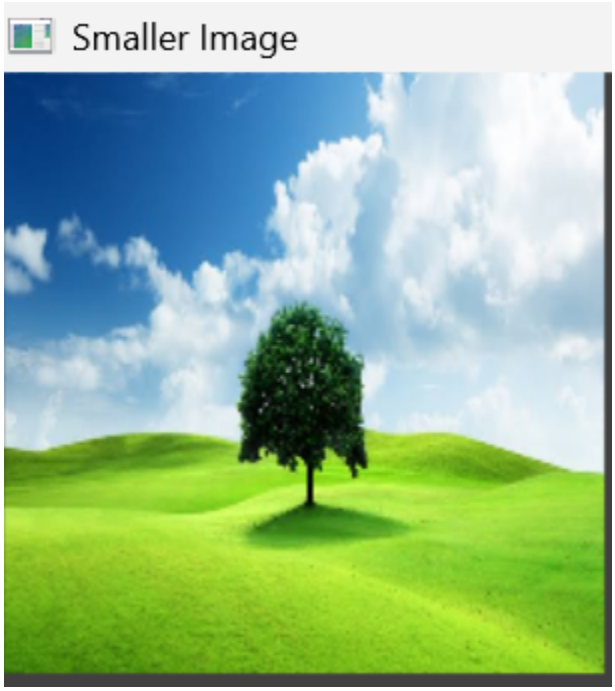
Bigger Image

600 × 600 pixels



Smaller Image

200 × 200 pixels



Result

Thus, image scaling was successfully implemented in Python using OpenCV to resize an image into both larger and smaller dimensions.